

A Multi-Arm A/B Testing and Causal Evaluation of Marketing
E-Mails
with Synthetic Validation and Heterogeneous Treatment Effects

Alexander Yu, Cameron Fong, Nathan Gin
Department of Statistics, University of California, Irvine

December 9, 2025

Contents

1	Introduction	4
2	Background and Motivation	4
3	Data Description	4
4	Exploratory Data Analysis	4
4.1	Treatment Assignment	4
4.2	Outcome Summaries	4
4.3	Covariate Balance	4
4.4	Correlation Structure	4
5	Causal Identification	4
5.1	Unconfoundedness	4
5.2	Overlap	4
5.3	SUTVA	4
5.4	Consistency	4
5.5	Why A/B Tests Identify the ATE	4
6	Synthetic Dataset Validation	4
6.1	Synthetic Data Generation	4
6.2	True Treatment Effects	4
6.3	ATE Estimator Comparison	4
6.4	Bias and Variance Evaluation	4
6.5	Interpretation	4
7	ATE Results on Hillstrom Dataset	4
7.1	Difference-in-Means	4
7.2	OLS / IPW Estimates	4
7.3	DR-Learner Results	4
7.4	Causal Forest Results	4
8	CATE and Heterogeneous Treatment Effects	4
8.1	CATE by Recency	4
8.2	CATE by Channel	4
8.3	CATE by Purchase History	4
8.4	Causal Forest Feature Importance	4
9	Discussion	4
10	Limitations	4
11	Conclusion	4
12	Additional Tables	4
13	Additional Figures	4

1	Introduction	
2	Background and Motivation	
3	Data Description	
4	Exploratory Data Analysis	
4.1	Treatment Assignment	
4.2	Outcome Summaries	
4.3	Covariate Balance	
4.4	Correlation Structure	
5	Causal Identification	
5.1	Unconfoundedness	
5.2	Overlap	
5.3	SUTVA	
5.4	Consistency	
5.5	Why A/B Tests Identify the ATE	
6	Synthetic Dataset Validation	
6.1	Synthetic Data Generation	
6.2	True Treatment Effects	
6.3	ATE Estimator Comparison	
6.4	Bias and Variance Evaluation	
6.5	Interpretation	
7	ATE Results on Hillstrom Dataset	
7.1	Difference-in-Means	
7.2	OLS / IPW Estimates	
7.3	DR-Learner Results	
7.4	Causal Forest Results	
8	CATE and Heterogeneous Treatment Effects	
8.1	CATE by Recency	
8.2	CATE by Channel	
8.3	CATE by Purchase History	